

Q1

② A stationary distribution: $\{X_n, n \geq 0\}$ is a Markov chain (mc) with state space $S = \{0, 1, 2, 3, \dots\}$ the mc is stationary iff: $[\pi_j^{(n)} = P(X_n = j) = P(X_{n+1} = j) = \pi_j^{(n+1)} \text{ for all } j \in S]$

Example is a two state chain with symmetric transitions:

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \Rightarrow \bar{\pi}_j^{(n)} = [0.5, 0.5]$$

NOTE: $\sum_{j \in S} P(X_n = j) = 1$
 $\sum \bar{\pi}_j = 1$

③ Limiting Distribution:

Suppose we have a transition matrix

$$P = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

Let X_n be the state at time n ; $X_0 = 1$

As we compute P^n for large n , the probability $P(X_n = j | X_0 = i)$ converges to a fixed value independent of the initial state.

Eventually

$$\lim_{n \rightarrow \infty} P(X_n = 1) = \bar{\pi}_1, \quad \lim_{n \rightarrow \infty} P(X_n = 2) = \bar{\pi}_2$$

where $\bar{\pi} = (\bar{\pi}_1, \bar{\pi}_2)$ satisfying $\bar{\pi}P = \bar{\pi}$ & $\bar{\pi}_1 + \bar{\pi}_2 = 1$

④ Periodic Markov Chain:

A state i has period d_i if any return to i must occur at multiples of d_i . A chain is periodic if all its states are periodic with the same d .

Ex: Two-state alternating chain

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{from state 1, must go to 2, then back to 1, alternating every 2 steps. } d(1) = 2, d(2) = 2 \rightarrow \text{Chain is periodic.}$$

⑤ Aperiodic Markov Chain:

A state i is aperiodic if it can return to itself at irregular intervals ($d_i = 1$).

Ex: All self-loops

$$P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \quad \text{from state 1, you can stay in 1 or go to 2. You can return to state 1 at times 1, 2, 3, ... } d(1) = 1.$$

⑥ Positive Recurrent Markov Chain:

A state j is positive recurrent if the expected return time to that state is finite:

$$m_j = E[T_j | X_0 = j] < \infty$$

{Theorem: if a Markov chain is finite & irreducible then all states are positive recurrent}

Ex: Ehrenfest model with 2 or 3 particles

- ① The number of particles in one side of the box evolves as a Markov chain
- ② finite state space $\rightarrow$ all states are positive recurrent
- ③ The Expected return time to any state is finite.

Null Recurrent Markov chain:

By def: A state i is null recurrent if:

- ① it is recurrent (you return eventually with probability 1)
- ② Expected return time is infinite $m_j = \infty$

Ex: Simple symmetric random walk on $\mathbb{Z}$ (Example 8.1.2)

- $\rightarrow$ state space: infinite
- $\rightarrow$ start at 0, go to ± 1 with equal probability
- $\rightarrow$ you return to 0 with probability 1, but Expected return time is infinite

$$P(2 < T_2 < 3 | T_1 < 2) = \frac{P(2 < T_2 < 3, T_1 < 2)}{P(T_1 < 2)}$$

$$P(2 < T_2 < 3 | T_1 < 2) = ?$$

Question 2 - Test 2

a) One step transition matrix

$$S = \{1, 2, 3\}$$

$$P = [P_{ij}], \text{ where } P_{ij} = P(X_{n+1} = j | X_n = i)$$

$$P_{ij} = \begin{pmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{pmatrix} = \begin{pmatrix} 0.6 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.3 \\ 0.0 & 0.1 & 0.9 \end{pmatrix}$$

b) $S = \{1, 2, 3\} \Rightarrow$ finite $\left\{ \begin{array}{l} \text{Every state is reachable from every other state} \\ \text{in some finite number of steps} \end{array} \right\} \Rightarrow$ Positive Recurrent.

$\{X_n, n \geq 0\} \Rightarrow$ irreducible. By standard Markov chain theory, if the Markov chain is irreducible, finite, & positive recurrent. Therefore, the Markov chain has a stationary distribution.

c) Solve for the stationary distribution:

$$\pi_j = \sum_i \pi_i P_{ij}, \quad \sum_j \pi_j = 1, \quad j, i \in S = \{1, 2, 3\}$$

$$\pi_i = \sum_j \pi_j P_{ji}, \quad i = 1, 2, 3 \text{ or } 1$$

$$\Rightarrow \pi_1 P_{11} + \pi_2 P_{21} + \pi_3 P_{31} \Rightarrow \pi_1 (0.6) + \pi_2 (0.2) + \pi_3 (0)$$

$$\pi_1 = 0.6\pi_1 + 0.2\pi_2 \Rightarrow \pi_2 = 2\pi_1$$

$$\pi_3 = \sum_i \pi_i P_{i3}, \quad i = 1, 2, 3$$

$$\Rightarrow \pi_1 P_{13} + \pi_2 P_{23} + \pi_3 P_{33} = \pi_3$$

$$\Rightarrow \pi_3 = 0.3\pi_2 + 0.9\pi_3 \Rightarrow \pi_3 = 3\pi_2$$

$$\sum \pi_j = 1 \Rightarrow \pi_1 + \pi_2 + \pi_3 = 1$$

$$\Rightarrow \pi_1 + 2\pi_1 + 6\pi_1 = 1$$

$$9\pi_1 = 1$$

$$\pi_1 = 1/9$$

$$\pi_2 = 2/9$$

$$\pi_3 = 6/9$$

$$\pi = (\pi_1, \pi_2, \pi_3) = (1/9, 2/9, 6/9)$$

d) Yes $\rightarrow$ If a Markov chain is irreducible, Aperiodic & positive recurrent, then the distribution is limiting distribution. The chain is irreducible & positive recurrent. $\{X_n, n \geq 0\}$ has a unique limiting distribution $\pi_{i,j}$. limiting distribution. $\pi(\pi_0, \pi_1, \dots, \pi_m)$ is the stationary distribution.

e) from (b): the chain is irreducible and the state space is finite. The chain is positive recurrent by definition of finite irreducible chains.

f) What is the period of the Markov chain?

$$\text{period } d(i) : d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}$$

Since the Markov chain is aperiodic ($d(i) = 1$) & the Markov chain is irreducible, & chain $\therefore$ Period of the chain is 1.

Question 3!

(a) Let X_n represent the number of deer on field B at time n
 $n = 0, 1, 2, 3, \dots$

State space: $S = \{0, 1, 2, 3, \dots, 2d\}$

Transition matrix:

$$P_{ij} = P(X_{n+1} = j | X_n = i) \text{ - values}$$

$$P_{i+1} = \frac{2d-i}{2d}$$

$$P_{i-1} = \frac{i}{2d}$$

$$P_0 = 0, \text{ otherwise}$$

(b) stationary distribution π

If a Markov chain has a stationary distribution

$$\pi_2(\pi_0, \pi_1, \pi_2, \dots)$$

$$\Rightarrow \pi_0 \sum_{i=0}^{2d} \pi_i P_{ij} = \pi_j \quad j \in S \Rightarrow \sum_j \pi_j = 1$$

$$\pi_0 = \sum_{i=0}^{2d} \pi_i P_{i0} \quad \boxed{i=1}$$

$$\Rightarrow \pi_1 P_{10} = \pi_0 = \frac{1}{2d} \Rightarrow \pi_0 = \frac{\pi_1}{2d} \quad \boxed{\pi_1 = 2d\pi_0}$$

$$\pi_1 = \sum_{i=0}^{2d} \pi_i P_{i1}, \quad i=2 \text{ or } 0$$

$$= \pi_0 P_{01} + \pi_2 P_{21}$$

$$\Rightarrow \pi_0 + \pi_2 \left(\frac{2}{2d}\right)$$

$$\pi_1 = \pi_0 + \pi_2 \left(\frac{2}{2d}\right) \Rightarrow 2d\pi_0 = \pi_0 + \pi_2 \left(\frac{2}{2d}\right)$$

$$2d\pi_0 - \pi_0 = \pi_2 \left(\frac{2}{2d}\right)$$

$$\pi_0(2d-1) = \pi_2 \left(\frac{2}{2d}\right) \Rightarrow \pi_2 = \frac{2d-1}{2} \pi_0$$

$$\text{for } j=2 \quad \pi_2 = \sum_{i=0}^{2d} \pi_i P_{i2} \quad i=0 \text{ or } 3$$

$$\Rightarrow \pi_1 P_{12} + \pi_3 P_{32}$$

$$\pi_2 = \pi_1 \left(1 - \frac{1}{2d}\right) + \pi_3 \left(\frac{3}{2d}\right)$$

$$\pi_3 \left(\frac{3}{2d}\right) = \pi_2 - \pi_1 \left(1 - \frac{1}{2d}\right)$$

$$\Rightarrow \frac{2d}{2} (n-1)\pi_0 - 2d\pi_0 \left(1 - \frac{1}{2d}\right)$$

$$\Rightarrow 2d\pi_0 \left(\frac{2d-1}{2} - \frac{2d-1}{2d}\right)$$

$$\Rightarrow 2d\pi_0 \left(\frac{2d(2d-1) - 2(2d-1)}{2d}\right)$$

$$\Rightarrow \pi_0 \left(\frac{(2d-1)(2d-2)}{2}\right)$$

$$\Rightarrow \pi_3 = \frac{2d(2d-1)(2d-2)\pi_0}{6}$$

$$\binom{N}{3} = \frac{2d}{3!} \frac{2d(2d-1)(2d-2)}{6} = \frac{2d(2d-1)(2d-2)}{6}$$

$$\Rightarrow \pi_{j-2} \left(\frac{2d}{j}\right) \pi_0 \quad j=0, 1, 2, \dots, 2d$$

$$\Rightarrow \sum_j \pi_j = 1 \Rightarrow \sum_{j=0}^{2d} \binom{2d}{j} \pi_0 = 1$$

$$\pi_0 = \pi_0 \sum_{j=0}^{2d} \binom{2d}{j} \quad j=0, 1, 2, \dots, 2d$$

① Is there a limiting distribution for the Markov chain?

Yes Because the chain is finite and irreducible and it is aperiodic.

$$\pi_j = \lim_{n \rightarrow \infty} P(X_n = j | X_0 = i) = \prod_{j=0,1,\dots,2d} \left(\frac{2d}{2d} \right)$$

Question 4:

② Let:

X_n represent the number of deer in field B at time step n

$n = 0, 1, 2, 3, \dots$ state space $S = \{0, 1, 2, 3, \dots, 2d\}$

Transition Probability:

$$P_{ij} = P(X_{n+1} = j | X_n = i) \quad i, j \in S$$

$$P_{i,i+1} = \frac{2d-i}{2d}$$

$$P_{i,i-1} = \frac{i}{2d}$$

$P_{i,0} = 0$, otherwise

Parameter = d

③ MLE

$$\vec{x}_T = (x_0, x_1, x_2, \dots, x_T)$$

$$n = 0, 1, 2, \dots, T$$

$$L(d | \vec{x}_T) = \prod_{n=0}^{T-1} P(x_{n+1} | x_n, d)$$

$$\Rightarrow \prod_{n=0}^{T-1} P(x_{n+1} = x_{n+1} | x_n = x_n, d)$$

Let

$$U = \{n : x_{n+1} = x_n + 1\}$$

$$D = \{n : x_{n+1} = x_n - 1\}$$

$$L(d | \vec{x}_T) = \left(\prod_{n \in U} \frac{2d - x_n}{2d} \right) \cdot \left(\prod_{n \in D} \frac{x_n}{2d} \right)$$

$$\ell(d) = \log L(d | \vec{x}_T) = \sum_{n \in U} \log \left(\frac{2d - x_n}{2d} \right) + \sum_{n \in D} \log \left(\frac{x_n}{2d} \right)$$

$$\Rightarrow \sum_{n \in U} [\log(2d - x_n) - \log(2d)] + \sum_{n \in D} [\log(x_n) - \log(2d)]$$

$$\text{let } T = |U| + |D|$$

$$\ell(d|\vec{x}_T) = \sum_{x_i \neq d} \log(2d - x_{i-1}) + \sum_{x_i = d} \log(x_{i-1}) - T \cdot \log(2d)$$

$$\hat{d}_{MLE} = \underset{d \in \max(x_{i-1} \text{ from } U)}{\text{Arg Max}} \ell(d|\vec{x}_T)$$

© $\vec{x} = (5, 6, 5, 6, 5, 7)$
 $x_0, x_1, \dots, x_4$

from $p_{\text{in}}(U)$ ↑
ups ↓
down

$$\ell(d) = \sum \log(2d - x_{i-1}) + \sum \log(x_{i-1}) - T \cdot \log(2d)$$

NOTE: we count the number of movement

$$\ell(d) = 3 \log(2d - 5) + 2 \log(6) - 5 \log(2d)$$

$$d \geq \frac{5}{2} = 2.5 \text{ from } 3$$

$$d=3 \approx -5.3753$$

$$d=4 \approx -3.5179$$

$$d=5 \approx -3.611$$

$$d=6 \approx -2.9003$$

$\ell(d)$ is max at $d=6$

$$\hat{d}_{MLE} = 6$$

Question 5:

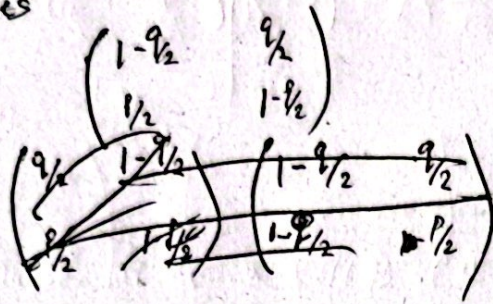
(a) Steady state distribution:

$$\Rightarrow \sum \pi_i p_{ij} = \bar{\pi}_j \text{ iies}$$

$$\Rightarrow \sum \bar{\pi}_i = 1$$

$$S = \{1, 2\}$$

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 1-q/2 & q/2 \\ q/2 & 1-p/2 \end{pmatrix}$$



$$\bar{\pi}_1 = \sum \bar{\pi}_i p_{i1}$$

$$\Rightarrow \bar{\pi}_1 p_{11} + \bar{\pi}_2 p_{21}$$

$$\Rightarrow \bar{\pi}_1 (1-q/2) + \bar{\pi}_2 (q/2) = \bar{\pi}_1$$

$$\Rightarrow \bar{\pi}_1 (q/2) + \bar{\pi}_2 (1-p/2) = \bar{\pi}_2$$

$$\Rightarrow \bar{\pi}_1 (1-q/2) + \bar{\pi}_2 (q/2) = \bar{\pi}_1 \Rightarrow \bar{\pi}_1 - \frac{q}{2}\bar{\pi}_1 + \frac{q}{2}\bar{\pi}_2 = \bar{\pi}_1$$

$$\Rightarrow -\frac{q}{2}\bar{\pi}_1 + \frac{q}{2}\bar{\pi}_2 = 0 \Rightarrow -q\bar{\pi}_1 + p\bar{\pi}_2 = 0 \Rightarrow p\bar{\pi}_2 = q\bar{\pi}_1 \dots \text{eqn (1)}$$

By Normalization $\bar{\pi}_1 + \bar{\pi}_2 = 1$

$$\text{from eqn (1)} \Rightarrow \bar{\pi}_2 = \frac{q}{p}\bar{\pi}_1$$

Sub in eqn (2)

$$\bar{\pi}_1 + \frac{q}{p}\bar{\pi}_1 = 1 \Rightarrow \bar{\pi}_1 \left(1 + \frac{q}{p}\right) = 1 \Rightarrow \bar{\pi}_1 = \frac{1}{1 + \frac{q}{p}} = \frac{p}{p+q}$$

$$\therefore \bar{\pi}_2 = \frac{q}{p} \cdot \frac{p}{p+q} = \frac{q}{p+q}$$

$$\boxed{\bar{\pi}_1 = \frac{p}{p+q}, \bar{\pi}_2 = \frac{q}{p+q}} \dots \text{Steady state distribution}$$

$$(b) E[T_2 | X_0 = 2] = \frac{1}{\bar{\pi}_2}$$

$$\text{from (a)} \bar{\pi}_2 = \frac{q}{p+q} \Rightarrow \therefore E[T_2 | X_0 = 2] = \frac{p+q}{q}$$

(c) The state space is finite (only 2 states) and chain is irreducible $\Rightarrow$ all states are positive recurrent. By Markov theorem, an irreducible positive recurrent chain

$$\Rightarrow \bar{\pi}_{22} = \frac{1}{E[T_2 | X_0 = 2]} = \bar{\pi}_2 = \frac{1}{\frac{p+q}{q}} = \frac{q}{p+q}$$